# Simple branch-point norm and multiplicity

[pcc:unverified] Hand derivation and prospective Lean source. Existing branch-stalk DVR and order-one source results are reused. No compiler, tests or trials were run.

### Simple branch: two valuation normalizations

Base: ord_a(X-a)=1 ⟶ Curve stalk: ord_P(x-a)=2, ord_P(y)=1

n(a)=0, b≠0: ord_P(n-by)=1 and ord_P(n+by)=1

Norm H=u1u2U: ord_P(H)=2 but ord_a(H)=1

Fixture $f=X^5-X$: curve support orders (2,0,0); base multiplicities (1,0,0)

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## Derivation on the actual stalk

At a rational simple branch P=(a,0) of $y^2=f(x)$, write $f=(X-a)q$ with $q(a)=f'(a)$ nonzero. The actual local ring is a DVR; y is a uniformizer and x-a is a unit times $y^2$. For n(a)=0 and b nonzero, $n=(X-a)m$, so $n\pm by=y(y\,q^{-1}m\pm b)$. Both parenthesized terms are units: each conjugate factor has order1 and their product order2. If n(a) is nonzero, both factors are units. For $H=n^2-b^2f=u1u2U$, the curve-stalk support-order sum is twice the selected-factor order. The base polynomial multiplicity sum is instead one: $H'(a)=-b^2f'(a)$ is nonzero. This does not double Mumford multiplicity. Two support factors vanishing at a would force b=0, outside the generic determinant guard.

## Actual coefficient fixture and countermodel

Over Q: $f=X^5-X$, a=0; $u1=X^2-X$, $u2=X^2+1$; both input ordinate polynomials0. Source coefficientData(-1,0,0,0,0,1,0,0)=(-2,0,0,-4,-4), hence n=0, b=-2, U=-4(X+1) and $n^2-u1u2U=4f$. Input divisibilities hold. Separability follows from $5f-Xf'=-4X$ and $f'(0)=-1$. The factors ±2y have order1; support germs have orders (2,0,0), base multiplicities (1,0,0). The naive nonbranch stalk formula would give 1=2.

## Source delivered and exact remaining work

Three prospective declarations: actual_branch_conjugate_orders; numerator_root_from_norm; two_support_roots_force_b_zero. They use the actual branch stalk and derive conclusions without assuming a desired valuation identity. No holes or new axioms are introduced; elaboration is unverified. The explicit missing formal bridge is $E(f)=Y^2$ for the branch evaluator, followed by mapping the polynomial norm identity to its stalk factorization. Base multiplicity transport requires the residual-unit criterion and finite factorization. Paper derivations are in DERIVATION.md; these bridges are not claimed as implemented. Infinity and global divisor assembly remain separate. Sources: existing WeightedProjFiniteBranchDVR, WeightedProjNormBranchZeroOrder and source actual-quintic modules; Mathlib derivative and valuation APIs; Stacks 00PD (DVR normalization), 0EXQ (ramification). Exact hashes and KB-unavailability record are in sources.json. Review, compilation and named axiom audit remain required.

Additional hand fixture (coordinator-suggested, independently substituted): $f=X(X^2-1)(X^2-4)$, $u1=X^2-1$,$u2=X^2-4$, zero ordinate polynomials yield coefficientData=(-9,0,0,-81,0), n=0,U=-81X. Points a=1,2,0 place the single simple base root in u1,u2,U respectively. Details: THREE-LOCATION-FIXTURE.md.